The cost of State Pension guarantees under forecast uncertainty*

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Abstract

Uprating rules that pay the highest of several indices each year have a fiscal cost that depends on the indices' joint path, not only on its mean. We study this for the adjusted Triple Lock, adopted in the UK's October 2026 Budget: from April 2030 the State Pension rises by the highest of 2.5%, CPI inflation and the amount needed to keep its April 2029 value relative to earnings. We show that its saving against the triple lock depends on which index leads each year and is zero on any path on which earnings always lead. We combine static microsimulation with stochastic paths of prices and earnings. The price–earnings model failed one condition of a test on past official forecasts registered before scoring, so we report a scenario envelope, not an expected value. In 2039-40 the rule lowers State Pension spending by £0.5bn (£0.3bn net of tax and benefits) on the Office for Budget Responsibility's (OBR) central forecast, £10.9bn (£7.3bn) on its long-term triple-lock assumption, and £11.4bn or £11.9bn averaged over 160 simulated paths with earnings growth 0.5 points lower or higher; the government estimates £15bn for Great Britain. A costing on one smooth path thus finds almost no saving. About 30% of households have lower income, and pensioner poverty after housing costs rises by up to 0.7 points.

Keywords: pension indexation; triple lock; path dependence; microsimulation; forecast evaluation; proper scoring rules; pre-analysis plans
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1 Introduction

Indexation rules set how pensions in payment keep pace with prices and earnings, and with them much of a pension system's long-run cost [Vidal-Meliá et al., 2009, OECD, 2023]. A rule that pays the highest of several indices each year is asymmetric: it passes on every year in which one index leads and keeps that gain in all later years. Its cost therefore depends on how variable the indices are and how often their order changes, not only on their average growth, and a costing on one central path, the convention in official medium-term forecasts, cannot show that dependence. This paper asks how large the dependence is for the fiscal and distributional effects of such a rule, and what can be reported when the model of the indices' joint path does not pass a test on past forecasts.

Since April 2011 the triple lock has raised the State Pension each April by the highest of prices, earnings growth and 2.5% (prices measured by RPI in its first year and by CPI since). Because it takes the highest of three figures every year, a year in which prices or the floor run ahead of earnings raises the pension for good, and later rises build on the higher level. The OBR's long-term projections assume that the triple lock raises the pension 0.6 percentage points a year faster than earnings [Office for Budget Responsibility, 2026c], and it estimates that by 2029-30 the triple lock will have added £15.5 billion a year to spending relative to earnings uprating since 2012 [Office for Budget Responsibility, 2025].

The government announced on 29 September 2026 that it would keep the triple lock until 2030 and then adjust it, and DWP published an analysis of the change the same day [Department for Work and Pensions, 2026b]. The Budget of 28 October 2026 adopted it as the adjusted Triple Lock: from April 2030 the State Pension rises "by the highest of 2.5%, inflation, or the amount required to maintain its value relative to earnings as reached in April 2029" [HM Treasury, 2026, para. 5.23 and Box 5.A]. The Budget adds that "It removes the 'ratchet' effect, whereby the State Pension rises ever higher relative to earnings" [HM Treasury, 2026, chapter 1], and puts the saving at: "Adjusting the Triple Lock is estimated to reduce State Pension spending by £15 billion a year by the late 2030s, and by £50 billion a year by the late 2040s" [HM Treasury, 2026, para. 1.29].

We use the adjusted Triple Lock as the application. It keeps the triple lock's protection against prices and the floor but removes the ratchet relative to earnings, so it isolates the part of the triple lock's cost that comes from the variability of the indices. We estimate its effect against the triple lock from 2027-28 to 2039-40 with full runs of PolicyEngine UK [Ahmadi et al., 2026] on the whole survey population, on the official central forecast, on the OBR's long-term assumption and on simulated paths of prices and earnings.

The paper makes three contributions. First, it gives an exact decomposition of the gap between the two rules (Section 3.2). The triple lock adds every year in which inflation or the floor beats earnings, like a ratchet; the adjusted rule follows a reflected random walk, a Lindley recursion [Lindley, 1952], whose gap from the triple lock depends on the running minimum of the cumulative difference between the indices, as a lookback option's payoff does [Goldman et al., 1979]. The gap never closes, is zero on any path on which one index always leads, and is bounded by the smaller of the two indices' cumulative leads, so the saving requires changes of lead. Second, it estimates the effects with static microsimulation [Sutherland and Figari, 2013], so that every figure includes the response of income tax, Pension Credit and other benefits, and it traces who bears the difference by income decile, household type and poverty status. Third, it tests the model of prices and earnings on past OBR forecasts with proper scoring rules [Gneiting and Raftery, 2007, Scheuerer and Hamill, 2015], under a pass rule and a reporting rule set before

scoring, in the spirit of pre-analysis plans [Casey et al., 2012, Olken, 2015]. The model failed one condition, and the paper shows what that leaves: a scenario envelope rather than an expected value.

On the OBR's central forecast the adjusted Triple Lock lowers State Pension spending by £0.5bn in 2039-40. On the OBR's long-term assumption about the triple lock the reduction is £10.9bn, and averaged over simulated paths with earnings growth moved 0.5 points either way it is £11.4bn to £11.9bn. The cost to households falls almost entirely on pensioners, with Pension Credit and income tax offsetting part of it.

Section 2 describes the triple lock's record and relates the paper to the literature. Section 3 states the two rules formally and derives where the saving comes from. Section 4 describes the data, the microsimulation, the paths of prices and earnings and the registered test. Section 5 gives the fiscal effects, the mechanism along single paths and the distributional effects. Section 6 compares the results with official costings, Section 7 discusses what they imply for costing such rules, and Section 8 concludes and states the main limitations.

2 Background and related literature

2.1 The triple lock since 2011

Figure 1 shows the rises actually paid in each April from 2011 to 2026. April 2011's rise, 4.6%, was set by RPI under that year's up-rating order [Social Security Benefits Up-rating Order, 2011], while the replay of the CPI rule gives 3.1%. From 2012, CPI set the rise five times, earnings six times and the floor four times. In April 2022 the law suspended the earnings leg, whose input was 8.4% on today's figures, so that year's rise was CPI, 3.1% [Social Security (Up-rating of Benefits) Act, 2021]. Over the 16 rises the pension rose by 89.4% as paid.

The rest of this subsection replays the rule on today's revised ONS figures, so that both rules see the same inputs. The replay differs from the rise paid in April 2016, 2019, 2020 and 2025, by 0.1 to 0.3 point, because earnings figures have since been revised, and in April 2011. On the replay the pension rose 87.4%, against 66.2% for earnings alone and 60.2% for prices alone, which leaves it 12.7% above where an earnings link alone would have taken it.

2.2 Indexation rules and path dependence

Automatic adjustment mechanisms link a pay-as-you-go pension system's benefits or contributions to indicators, so that its finances respond to demographic and economic change without new legislation [Vidal-Meliá et al., 2009]. The indexation of pensions in payment is one such link; the OECD's review of the inflation surge since 2021 records how far different rules protected pensioners' purchasing power [OECD, 2023]. The triple lock is unusual in taking the highest of three figures each year, and Portes [2023] places it in the longer history of the State Pension's value relative to earnings. A rule that pays the higher of two growth rates and compounds the result has the structure of a path-dependent option: each year adds the positive part of the difference between the indices, and the adjusted rule's return to its earnings path makes the gap between the rules depend on the running minimum of the cumulative difference, as the payoff of a lookback option depends on the running minimum of a price [Goldman et al., 1979]. We use the analogy for the structure of the payoff only; nothing in the paper prices an option.

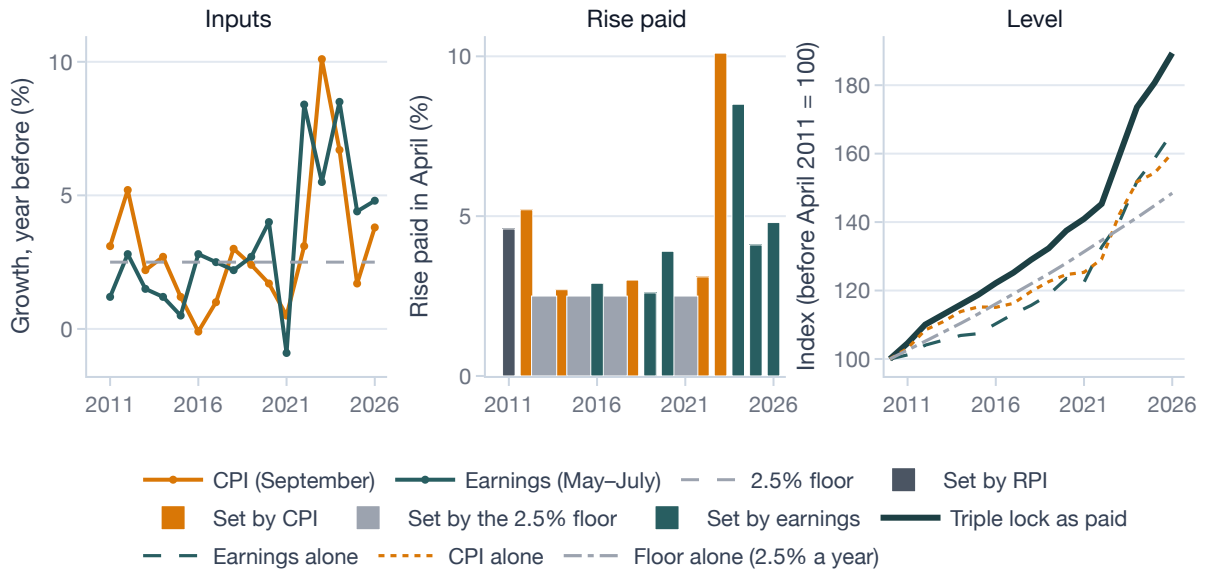


Figure 1: The triple lock since 2011: the rises paid, and a level that ended well above earnings alone. Left: September CPI and May–July earnings growth in the year before each April (today's ONS figures), and the floor. Centre: the rise paid, coloured as the input that set it. Right: the level as paid, against earnings alone, CPI alone and the floor alone (2.5% a year).

2.3 Uncertainty in long-run pension costings

Long-run projections of pension spending are uncertain, and stochastic projections of social security finances express that uncertainty as distributions over paths of demographic and economic inputs [Lee and Tuljapurkar, 1998]. The OBR's fiscal risks and sustainability reports treat the triple lock as a source of fiscal risk and project its cost against earnings uprating [Office for Budget Responsibility, 2025]. Central banks publish fan charts to show the uncertainty around a central projection [Britton et al., 1998], and official forecasts have been found biased, with larger bias at longer horizons [Frankel and Schreger, 2013]. Two points carry over. The cost of a rule that is convex in its inputs should be evaluated over a distribution of paths, not on the central one; and a model that supplies the distribution should be tested against outcomes, for which proper scoring rules reward a forecaster for reporting its true predictive distribution [Gneiting and Raftery, 2007], including for multivariate outcomes [Scheuerer and Hamill, 2015].

2.4 Microsimulation of pension policy

Static tax-benefit microsimulation applies the rules of the tax and benefit system to a representative survey of households, with behaviour held fixed, and is the standard tool for the distributional analysis of benefit reforms [Sutherland and Figari, 2013]; UKMOD is one such model for the four nations of the UK [Richiardi et al., 2021]. Dynamic microsimulation models instead project individual life courses, including pension entitlements, forward in time [Li and O'Donoghue, 2013]; DWP's costing of the adjusted Triple Lock uses one (Section 6). Our runs are static: they reweight today's population to projected totals and recalculate every tax and

benefit under each rule.

2.5 Registered analysis plans

Pre-analysis plans fix a study's hypotheses and tests before the data are seen, so that results cannot be chosen after the fact [Casey et al., 2012, Olken, 2015], and Christensen and Miguel [2018] review their spread in economics. They are rare in model-based policy costing, where the analyst chooses both the model and its evaluation. We registered the forecast-evaluation test and the rule that follows from its outcome, including what the paper reports if the model fails. One element of the test was changed after the first scores were seen; Section 4.4 reports the change and its reason.

3 Policy rules

3.1 The two rules

Under the triple lock the full basic and new State Pension rise each April by the highest of three figures: the CPI 12-month rate for the previous September [Office for National Statistics, 2026b]; growth in average weekly earnings, whole-economy total pay, in May to July of the previous year [Office for National Statistics, 2026d]; and 2.5%. ONS publishes both inputs to 0.1 point, and both rules here use them at that precision.

The adjusted Triple Lock follows the triple lock to April 2029. From April 2030 let L_t be the pension's level and A_t an earnings path that starts at the 2029-30 level and grows with May–July earnings, $A_t = A_{t-1}(1 + e_{t-1})$. Then

$$L_t = \max\big(L_{t-1}\,(1 + \max(c_{t-1}, f)),\ A_t\big),$$

where c is September CPI, e is May–July earnings growth and f is the 2.5% floor. The pension never rises by less than the higher of CPI and the floor, and never falls below its earnings path. In April 2030 this rise equals the triple lock's, since the earnings path starts at the same level, so the two rules can first differ in April 2031 and the saving in 2030-31 is zero on every path. As a modelling assumption we round any top-up to the earnings path up to the next 0.1 point, so the pension never ends below that path; this favours the pensioner by at most 0.1 point a year.

The Budget's wording fixes the anchor: the pension's value relative to earnings "as reached in April 2029". DWP's description says the same in other words: the pension will "reach a record high relative to earnings over the coming three years", and from April 2030 it keeps pensioners benefiting "via the State Pension retaining that record high value relative to earnings": "In any given year it will go up by at least inflation or 2.5% – and anything more that is needed to retain that value". It adds: "This means that over time the State Pension is set to rise in line with average earnings" [Department for Work and Pensions, 2026b]. Two choices remain ours. The rule uses the triple lock's measures, September CPI and May–July earnings, and the top-up is rounded up, as above.

The law requires only that the basic and new State Pension rise at least in line with earnings [Social Security Administration Act, 1992]; the prices leg and the floor are policy on top. In a year when the adjusted Triple Lock rises by the higher of CPI and the floor while its level returns towards its earnings path, its rise is below earnings growth, which section 150A does not allow, so the rule would need primary legislation, as the suspension of the earnings leg for

April 2022 did [Social Security (Up-rating of Benefits) Act, 2021]. On both OBR scenarios this happens once, in April 2032; on the two simulated paths shown in Section 5.4, in four and five of the Aprils from 2031 to 2039.

3.2 A formal statement

Write $m_t = \max(c_{t-1}, f)$ for the higher of CPI and the floor that applies in April t, and T_0 for April 2029, the last rise under the triple lock alone. With P_t the triple lock's level and L_t and A_t as above,

$$P_t = P_{t-1}\,\big(1 + \max(m_t, e_{t-1})\big), \qquad L_t = \max\big(L_{t-1}\,(1+m_t),\ A_t\big), \qquad A_t = A_{t-1}\,(1+e_{t-1}),$$

with $P_{T_0} = L_{T_0} = A_{T_0}$. Let $d_t = \ln(1+m_t) - \ln(1+e_{t-1})$, positive in a year in which CPI or the floor beats earnings, and measure each rule's level against the earnings path, $y_t = \ln(P_t/A_t)$ and $x_t = \ln(L_t/A_t)$. Then

$$y_t = y_{t-1} + \max(d_t, 0), \qquad x_t = \max(x_{t-1} + d_t,\ 0), \qquad y_{T_0} = x_{T_0} = 0.$$

The triple lock accumulates every positive d_t, while the adjusted rule's x_t is a cumulative sum reflected at zero, a Lindley recursion [Lindley, 1952]. With $D_t = \sum_{s=T_0+1}^{t} d_s$ and $D_{T_0} = 0$, the log gap between the rules, $G_t = y_t - x_t$, is

$$G_t = \sum_{s=T_0+1}^{t} \max(-d_s, 0) + \min_{T_0 \le s \le t} D_s,$$

which satisfies

$$0 \le G_t \le \min\Big(\sum_{s=T_0+1}^{t} \max(d_s, 0),\ \sum_{s=T_0+1}^{t} \max(-d_s, 0) \Big).$$

Three properties follow. The gap never falls, $G_t \ge G_{t-1}$, so a rise the triple lock keeps is a saving in every later year. The gap is zero on any path on which earnings lead in every year, and on any path on which they never lead, so a saving requires a change of lead. And because $G_t \ge 0$ on every path, a smooth central path on which earnings lead every year gives the lowest possible saving, and any variation in the inputs that produces a change of lead raises the average saving above it.

The gross saving in a fiscal year is the difference in basic and new State Pension paid,

$$S_t^{g} = \sum_i w_{it}\,\big(b_{it}^{\mathrm{TL}} - b_{it}^{\mathrm{A}}\big) \approx \big(1 - e^{-G_t}\big)\,B_t^{\mathrm{TL}},$$

where w_{it} is the survey weight of person i, b_{it} their basic or new State Pension under each rule and B_t^{TL} total spending on both under the triple lock; the approximation is exact, up to rounding, when each person's amount is a fixed share of the full rate. The net saving is the change in the government's direct budget balance, $S_t^{n} = \sum_h w_{ht}\,(\tau_{ht}^{\mathrm{A}} - \tau_{ht}^{\mathrm{TL}})$, where w_{ht} is the weight of household h and τ_{ht} the direct taxes it pays less the benefits it receives, State Pension included. The rounding of the top-up to the earnings path adds a small non-negative term to x_t, so the identities hold up to that rounding in the model runs.

3.3 Where the saving comes from

When earnings lead, both rules give the same rise. The rules part only after a year in which the higher of CPI and 2.5% beats earnings. Both pay that higher figure, but it lifts the adjusted Triple Lock above its earnings path. When earnings lead again, the triple lock rises with earnings from the higher level, while the adjusted Triple Lock rises only by the higher of CPI and the floor until its earnings path catches up. Under the triple lock each such rise stays in the pension's level; under the adjusted rule the level returns to its earnings path over the following years.

A two-year example shows the arithmetic. Suppose CPI is 6% and earnings growth 2% in the first year, then CPI 2% and earnings 4%. The triple lock rises by 6% and then 4%, 10.24% in all. The adjusted Triple Lock also rises by 6% in the first year, which leaves it above its earnings path, and then by the floor, 2.5%, 8.65% in all. It ends 1.4% below the triple lock (1.59 points less cumulative growth).

So the saving is zero on any path on which earnings lead in every year after the switch, and positive on a path on which the higher of CPI and the floor leads in some year and earnings lead again later. It depends on how often the lead passes between earnings and the higher of prices and the floor, not only on their average growth. The triple lock compounds each year's highest input, so its level over uncertain inputs exceeds its level on the average inputs (Jensen's inequality). The adjusted rule's level is tied to an earnings path that does not compound those highs. A smooth central path has few changes of lead by construction; simulated paths, like past years, have many more (Section 4). Section 3.2 states this formally.

4 Data and method

4.1 Model and data

Every saving and household figure for a given path is a full run of PolicyEngine UK [Ahmadi et al., 2026, PolicyEngine, 2026b,c] on the whole survey population, once under each rule. PolicyEngine UK is a static tax-benefit microsimulation model of the kind Sutherland and Figari [2013] describe: it applies the tax and benefit rules of each year to survey households and allows no behavioural response. All runs use one release bundle: policyengine 6.11.1, which pins policyengine-uk 2.151.0 and policyengine-core 3.33.3, with the national Microcosm dataset for 2024-25 released with it (78,234 households) [PolicyEngine, 2026a]. Constituency and local-authority runs use the same release's local-area cut. The Enhanced Family Resources Survey, release 1.58.2, runs only for the check against DWP's caseloads (Table 8). A path's CPI and earnings growth replace the model's economic assumptions before the data load, so benefit rates, tax thresholds and earnings follow it, and every run checks that they do. Only the basic and new State Pension differ between the two runs of a path. Under both rules the additional State Pension rises with September CPI and State Pension age follows the law's timetable. Money is in cash terms for the UK unless stated.

Three conventions shape the net saving. First, the Pension Credit guarantee rises with May–July earnings, the minimum section 150A requires, under both rules. If it instead rose by the same percentage as the State Pension under each rule, it would be lower under the adjusted rule, the Pension Credit offset would be smaller and the net saving larger. Second, the income tax personal allowance stays at £12,570 to 2030-31, as current law sets it, and then rises with each path's September CPI; the full new State Pension is above it throughout, so a lower pension means less income tax even for a pensioner with no other income. Third, Pension Credit take-

up follows the dataset's own claim decision for each benefit unit, so a household newly entitled under the adjusted rule claims only if that decision says it would; Housing Benefit and council tax reduction respond only for households already receiving them, and passporting from the guarantee credit follows actual receipt.

4.2 The population in each year

The dataset represents households in 2024-25, and each record stands for a person of its age in every year. Through 2025, the year the dataset calibrates to, each record keeps its own weight; after it, household weights are reweighted so that each age and sex group grows as in the ONS's 2024-based population projection [Office for National Statistics, 2026e]. Ages are not top-coded, so every recorded age is kept. Each person's State Pension type follows from their date of birth: those who reached State Pension age before 6 April 2016 are on the basic State Pension, later cohorts on the new one [Pensions Act, 2014]. Deaths, migration and contribution histories are not simulated, and pensioners abroad are outside the dataset. Records carried forward from today's pensioners may understate the entitlements of people reaching State Pension age in the 2030s, most of whom will have spent their whole working lives under the new State Pension.

4.3 Paths of prices and earnings

The simulated paths come from a model of monthly changes in the CPI index [Office for National Statistics, 2026a] and in average weekly earnings [Office for National Statistics, 2026c], fitted to data from 2000 to the latest published month, with the furlough months of 2020–21 left out. Each month's change depends on the previous month's changes in both series, with fixed coefficients; the shocks are past pairs of residuals drawn at random, so prices and earnings keep their joint variation, but the model has no periods of higher or lower volatility and no uncertainty about its coefficients. From the simulated months come both the statutory inputs (September CPI, May–July earnings growth) and the calendar-year growth the rest of the model uses. Every path is then shifted by the same smooth drift so that the average matches the OBR's forecast in every year, while each path keeps its own ups and downs. The earnings scenarios move that drift by 0.5 points a year from 2031. In the inputs that set the April rises of 2011 to 2026, with April 2022 as in law, the lead between CPI and earnings changed 0.33 times a year.

4.4 Testing the model on past forecasts

An average over the simulated paths needs a model whose paths behave like the real inputs, so we registered a test on past OBR forecasts [Office for Budget Responsibility, 2026b] before scoring any model. Five versions of the model of prices and earnings are fitted only on data available before each forecast, centred on that forecast, and scored, with proper scoring rules for each statistic and for the joint outcome [Gneiting and Raftery, 2007, Scheuerer and Hamill, 2015] and with the coverage of central bands, on the gap between earnings and CPI in each year, on how often the lead changes and the floor binds, and on the gap the two rules open after four rises. One test uses seven forecasts, one a year from the March 2016 EFO to the March 2022 EFO; another uses thirteen, from the June 2010 Budget forecast to the March 2022 EFO. Table 1 gives the pass rule.

Under the first rule every version failed, because April 2022 was scored on the published earnings: each had only one of six four-year outcomes inside its band, because the furlough schemes distorted 2021 pay growth. We then changed that one element, before scoring again,

to score April 2022 as the law applied it, with the earnings leg suspended; the rest of the rule is unchanged, and the change was made after seeing scores. The second rule also fixed what follows a failure: only the main version may supply an average, no other version may be promoted on its scores, and if the main version fails the release reports scenarios only.

On the Budget's inputs the main version failed one condition (Table 9). Its average error in the share of years in which earnings growth falls below the floor was 0.267 on the seven-forecast test, against a limit of 0.25: it expected earnings below 2.5% in 44.6% of the years tested, against 17.9% in practice. It passed every other condition: the outcome fell inside its central 80% band in five of seven four-year windows and 83% of annual checks. Two other versions passed and two failed on the same condition, but the rule forbids promoting another version, so the release carries no expected value and this paper reports the scenario envelope only. The test has little power. With seven overlapping four-year windows, a model whose bands cover far less than 80% could still pass, so passing would show only that the model is not clearly wrong; failing one condition by 0.017, less than that statistic's standard error of 0.085, does not show the model is badly wrong either.

Table 1: The pass rule for the test on past forecasts, April 2022 scored as the law applied it.

Condition	Threshold
Coverage	Between 50% and 100% of outcomes, yearly and after four years, fall inside the central 80% band
Coverage, plausible range	The range of plausible coverage after four years includes 80%
Bias in the yearly gap	Average error in earnings minus CPI at most 1.5 points
Bias in lead changes	Average error in the number of lead changes at most 1.0 a forecast
Bias in floor years	Average error in the share of years in which the floor binds at most 0.25
Accuracy scores	No score more than 25% worse than the main version's
Past years	The realised gap since 2011 between the 5th and 95th percentiles of its paths

Note: A version passes only if it meets every condition on both tests.

4.5 Future releases

A third registered rule governs releases that use OBR forecasts published after 28 October 2026; this release does not use it. Each such release publishes two averages of the main version's paths, one with equal weights and one recalibrated so that each year's share of paths with earnings growth below 2.5% matches the model's past error on that statistic. Both carry the label model-conditional sensitivity and sit beside the scenario envelope; neither may be called central, expected or a forecast. A prospective test on outcomes nobody has yet seen, from determination year 2027, decides only whether the label may become model-conditional estimate, and it gives no verdict until eight determination years have joined the test, so not before October 2034, when September 2034 CPI appears. On simulated outcomes a correct model passes 35.8% of the time after four years, 68.5% after eight and 85.6% after twelve, and a model that states double the

true rate of earnings growth below the floor passes 35.1%, 43.1% and 63.7%: four years cannot tell the two apart.

5 Results

5.1 Fiscal effects on four scenarios

Table 2 and Figure 2 give the reduction in State Pension spending under the adjusted Triple Lock, relative to the triple lock, on four scenarios. Two are single full runs. One is the OBR central forecast. The other, the OBR long-term path, keeps the central forecast's inputs but pays the triple lock at the OBR's long-term triple-lock rate from April 2032, which phases in to earnings growth plus 0.6 points, and applies the adjusted rule to the same inputs. The other two take 160 simulated paths of prices and earnings, each a full run under both rules, and move calendar-year earnings growth 0.5 points a year lower or higher from 2031; each figure is the average over those paths, and its margin is the 95% Monte Carlo error from sampling them. We attach no probability to any of them, and the four do not form a probability interval: the registered rule withdrew the average over the unshifted paths (Section 4.4).

On the OBR long-term path the saving is £10.9bn in 2039-40, £7.3bn once income tax and benefits respond: extra Pension Credit offsets £0.8bn and lower income tax £2.4bn. In this scenario earnings lead CPI and the floor in every April from 2032, so the adjusted Triple Lock returns to its earnings path and then follows it, and the whole of the OBR's premium counts as saving. The premium is the OBR's average allowance for years in which CPI or the floor beat earnings; no path of CPI and earnings pays it every year, so this run is a comparison with the OBR's costing convention, not a forecast. On the two earnings scenarios it is £11.4bn and £11.9bn. Higher earnings growth raises the saving through the mid-2030s, £4.4bn against £3.1bn in 2034-35, and by 2039-40 the two are about £0.5bn apart.

Table 2: Reduction in State Pension spending under the adjusted Triple Lock relative to the triple lock on four scenarios, £ billion, UK, cash terms.

	2034-35		2039-40	
Scenario	Gross	Net	Gross	Net
OBR central forecast	0.4	0.3	0.5	0.3
OBR long-term path (triple lock at the OBR's long-term rate from April 2032)	3.1	2.1	10.9	7.3
Earnings −0.5 points from 2031, average of 160 paths	3.1 ± 0.4	2.2 ± 0.3	11.4 ± 0.5	7.8 ± 0.3
Earnings +0.5 points from 2031, average of 160 paths	4.4 ± 0.5	3.0 ± 0.3	11.9 ± 0.5	8.1 ± 0.3

Note: The OBR central forecast and the OBR long-term path are single full runs. The earnings scenarios are averages over 160 simulated paths, each run in full, ± the 95% Monte Carlo error from sampling the paths. A scenario envelope, not a probability interval.

5.2 Why the OBR's central forecast yields a small saving

The OBR's central forecast is one smooth path. On it earnings beat CPI and the floor in almost every year after the switch, so the two rules give nearly the same rise: the two first differ in April 2032, and the saving is £0.4bn in 2034-35 and £0.5bn in 2039-40 (£0.3bn net). The

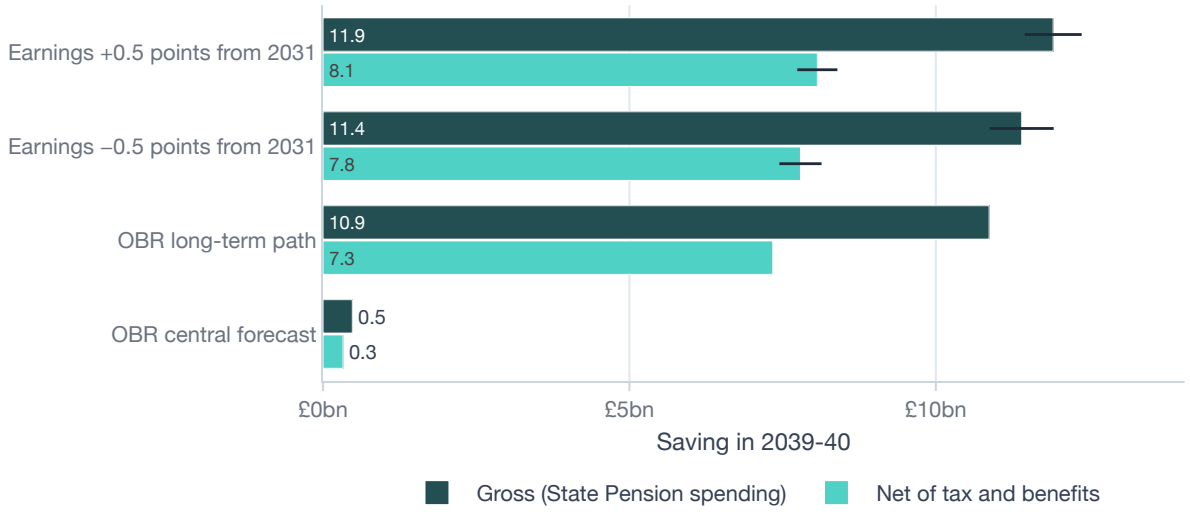


Figure 2: The saving depends on the path: small on the OBR's smooth central forecast, far larger where prices and earnings swap the lead or the triple lock outpaces earnings. Gross and net saving in the final year on each scenario, £ billion; whiskers are the 95% Monte Carlo error on the draw averages.

OBR finds the same for its forecast period: "in our current medium-term forecast an uprating of 2.5 per cent would apply under both the previous and adjusted version of the triple lock in both 2030-31 and 2031-32" [Office for Budget Responsibility, 2026a, para. 3.50]. Real years are bumpier. Whenever inflation spikes or earnings growth dips below the floor, the triple lock keeps the higher rise for good while the adjusted rule's level later returns to its earnings path, so the saving grows. The saving is never negative, and the central path has fewer such years than the simulated paths, so its saving is far below theirs.

5.3 The mechanism on the OBR central forecast

The central path uses the OBR's October 2026 Economic and fiscal outlook, published with the Budget, for calendar years to 2030 [Office for Budget Responsibility, 2026a], and the OBR's March 2026 long-term economic determinants after that [Office for Budget Responsibility, 2026c], which the OBR describes as a projection, not a forecast; the OBR published no long-term determinants with its October forecast. April 2027's State Pension rise uses the published May–July 2026 earnings growth, 3.9%, and September 2026 CPI, 3.5% [Office for National Statistics, 2026d,b]. For the Aprils 2028 to 2031 the inputs are the OBR's September-quarter CPI and April–June earnings growth; later Aprils use its calendar-year figures. The OBR forecasts average earnings on a national-accounts measure, which stands in here for May–July average weekly earnings. Both rules share these inputs.

The floor sets the triple lock in April 2029, 2030 and 2031; from April 2032 earnings lead every year (Figure 3). The two rules differ in one year, April 2032: after the floor years the adjusted Triple Lock rises only by the floor, or by what brings it back to its earnings path, while the triple lock rises with earnings from the higher level. By 2039-40 the full new State Pension is £18,876 a year under the triple lock and £18,840 under the adjusted rule, and the saving is

£0.5bn (£0.3bn net).

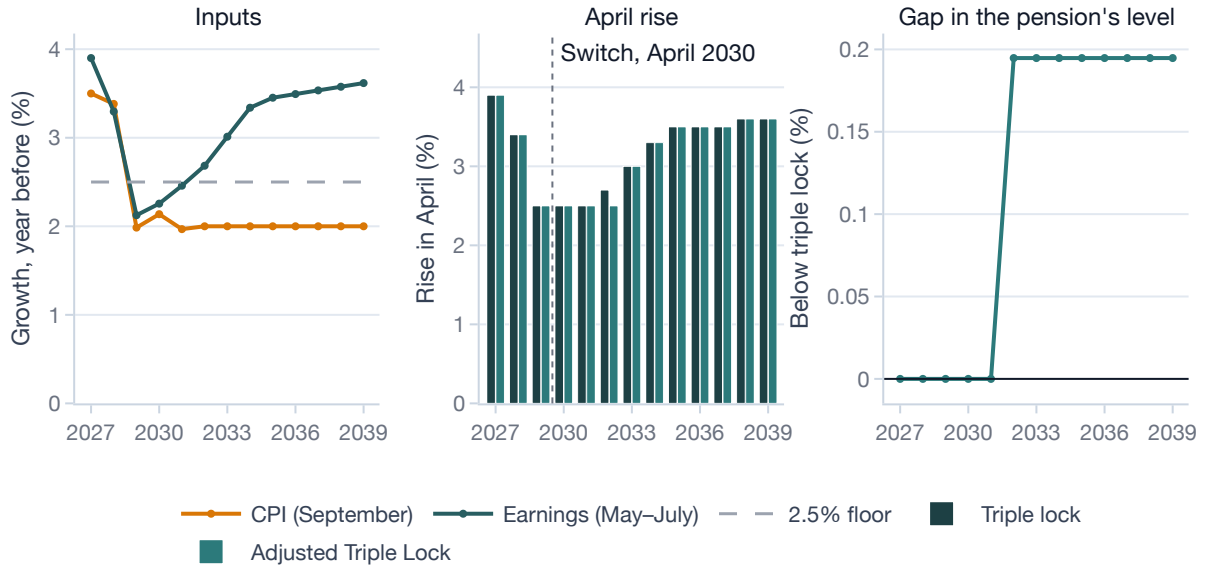


Figure 3: On the OBR central forecast the floor sets the early rises, and the adjusted Triple Lock differs from the triple lock in only a few Aprils. Left: the inputs behind each April's rise, shown at the April they set. Centre: the April rise under each rule; the dotted line marks the switch. Right: how far the adjusted rule's full new State Pension sits below the triple lock's.

5.4 The mechanism on simulated paths

The simulated paths come from a monthly model of prices and earnings, shifted so that their calendar-year averages match the OBR's forecast (Section 4). The results file runs three of them in full: one drawn at random, and two picked for the size of the gap between the rules they open by 2039-40, one middle-sized and one large. Each is one path the model makes, not a forecast, and carries no probability weight. Figure 4 shows how far apart the two rules end on each, beside the two OBR scenarios, and Table 3 the saving on each. On the path with a middle-sized gap the saving is £11.3bn in 2039-40 (£7.7bn net), close to the OBR long-term path; on the path with a large gap it is £19.8bn. Most of a path's saving comes from a few years of high inflation or weak earnings, after which the gap stays.

5.5 A historical counterfactual

The same comparison can be run backwards on the replayed inputs. Had the adjusted Triple Lock started in April 2012, 2013, 2014 or 2015, its level in 2026-27 would be 6.2% below the triple lock's on the same inputs (Figure 5). Applied to the full new State Pension actually paid, £12,548 a year, that would be £11,769. Most of the difference follows April 2023, when September CPI of 10.1% set a rise far above earnings growth of 5.5%. Started in April 2024, 2025 or 2026, the rule would have paid the same as the triple lock every year. In full model runs, State Pension spending in 2026-27 would have been £7.8bn lower, and the government

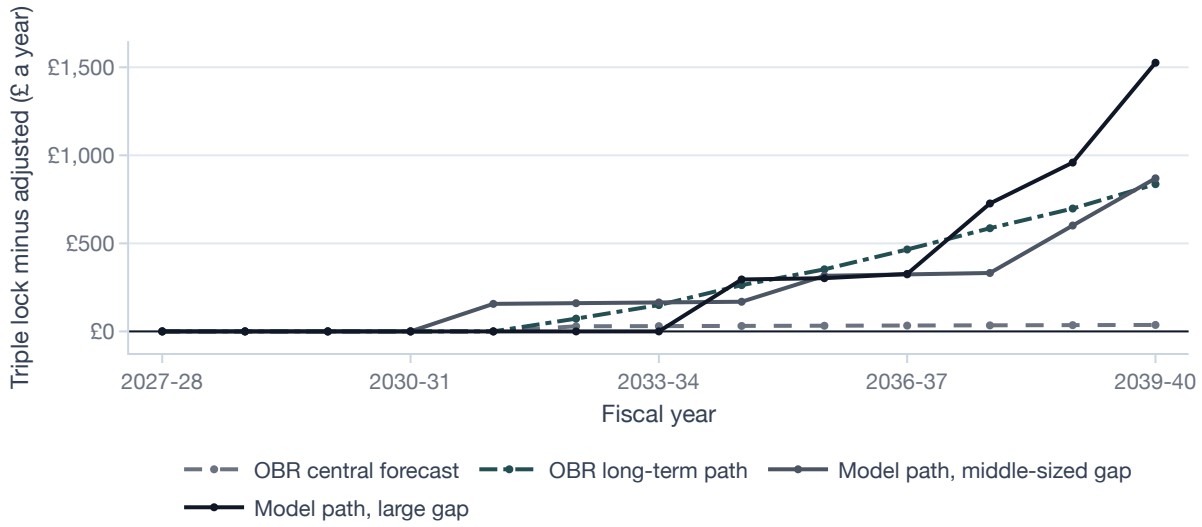


Figure 4: On simulated paths the gap opens in a few steps and then persists. How far the adjusted Triple Lock's full new State Pension ends below the triple lock's on each path shown, £ a year.

Table 3: The paths run in full, in the final year: the full new State Pension under each rule, the saving and the average loss among households with less income than under the triple lock.

Path	Triple lock (£ a year)	Adjusted (£ a year)	Gross (£bn)	Net (£bn)	Households with less income	Average loss (£ a year)
OBR central forecast	£18,876	£18,840	0.48	0.33	30%	£35
OBR long-term path	£19,676	£18,840	10.87	7.34	30%	£801
Random model path	£21,045	£20,028	13.23	9.06	30%	£973
Model path with a middle-sized gap	£18,782	£17,912	11.30	7.69	30%	£836
Model path with a large gap	£19,630	£18,104	19.83	13.45	30%	£1,459

Note: On every path about the same share of households has less income.

£5.6bn better off once tax and benefits respond. Started in April 2017, a decade before the last rise, it would leave the pension 5.1% lower by 2026-27. The OBR replays the rule on outturn from 1990 to 2025: "Averaging over the period as a whole, state pension payments would have been on average 1.4 per cent lower each year than under the current triple lock and 4.4 per cent higher than under earnings uprating" [Office for Budget Responsibility, 2026a, para. 3.51].

5.6 Distributional effects

The household results run each path through PolicyEngine UK for the whole survey population, once under each rule. Every pensioner's State Pension is recalculated, and with it their income tax, Pension Credit, Housing Benefit, council tax reduction and Winter Fuel Payment. We report them on the OBR long-term path, the largest of the single runs, and give the poverty changes on the OBR central forecast and the two simulated paths beside it. Cash and percentage changes in income scale roughly with the gap in the pension; poverty rates do not.

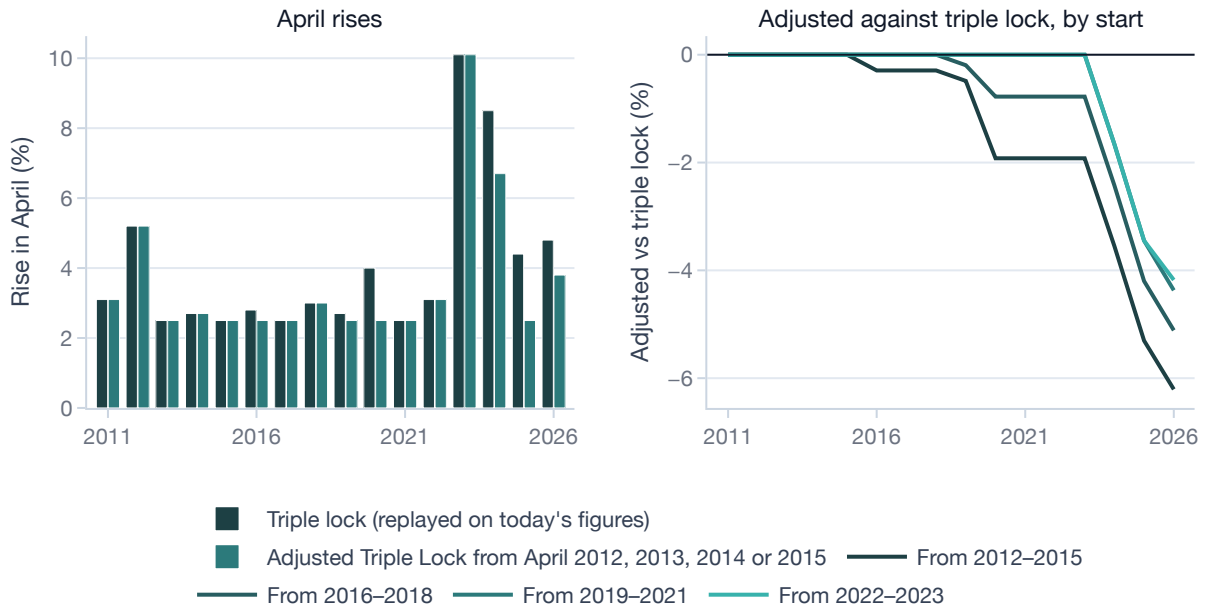


Figure 5: Had the adjusted Triple Lock started earlier, the pension would now be lower, mostly because of one inflation spike. Left: the April rises under the triple lock and under the adjusted rule from the earliest start that changes anything, both replayed on today's figures. Right: the adjusted rule's level against the triple lock's, by start year.

5.7 From gross to net

Table 4 follows the gross saving on the OBR long-term path to the net saving. In 2039-40 extra Pension Credit offsets £0.81bn of the £10.87bn gross saving and lower income tax £2.39bn, leaving £7.34bn net, 68% of the gross. The net figure counts direct taxes and benefits only, with no change in behaviour. It leaves out VAT and other indirect taxes on pensioners' lower spending, which would make it smaller. Not every offset falls on the same budget: council tax reduction is paid by local authorities, and part of the income tax would otherwise go to the Scottish and Welsh governments.

5.8 Who has less income

On every path about 30% of UK households have at least £1 a year less income under the adjusted Triple Lock than under the triple lock in 2039-40: the share is set by who receives a State Pension, while the size of the loss depends on the path (Table 3). On the OBR long-term path those households have £801 a year less on average, against £35 on the OBR central forecast. On the OBR long-term path single pensioners have 0.8% less income than under the triple lock on average, and pensioner couples 1.3%.

Figure 6 ranks households by equivalised household net income before housing costs under the triple lock, with each decile holding a tenth of people, and gives each decile's total change in income as a share of its total income. Relative to income, the change is largest in the 3rd decile, at 0.56% of income, and smaller in the lowest-income and in the higher-income deciles.

Table 4: From gross to net on the OBR long-term path in the final year, £ billion.

	2039-40 (£bn)
Gross saving (State Pension)	10.87
Extra Pension Credit	−0.81
Extra Housing Benefit	−0.23
Extra council tax reduction	−0.05
Extra Winter Fuel Payment	−0.03
Less income tax	−2.39
Other taxes and benefits	−0.03
Net saving	**7.34**

Note: Each offset is signed so the column adds up to the net saving.

The guarantee credit, which makes up a lower pension pound for pound, and the State Pension's smaller share of higher incomes both work in this direction.

On the OBR central forecast, where the pension gap is small, no poverty measure changes by more than 0.03 percentage points. On the OBR long-term path pensioner poverty after housing costs is 0.70 points higher on the absolute measure and 0.75 points higher on the relative one, and on the simulated path with a large gap up to 1.52 points higher (Table 5). Poverty among everyone changes much less. The absolute line is 60% of the 2024-25 median, held fixed in real terms by uprating it with each path's CPI; the relative line is 60% of median income in each run, so it is itself a little lower under the adjusted rule. Pension Credit makes up much of a lower pension for pensioners on the guarantee, but not for those just above it.



Figure 6: The loss is largest, relative to income, in the lower-middle deciles. Each decile's total change in household net income under the adjusted Triple Lock against the triple lock, as a % of its total income, on the OBR long-term path in the final year, by decile of equivalised household net income before housing costs under the triple lock (each decile holds a tenth of people).

Table 5: Poverty after housing costs in the final year: share of people in households below the line, %, and the change under the adjusted Triple Lock, percentage points.

Measure	Triple lock (%)	Adjusted (%)	Change, OBR central (points)	Change, OBR long term (points)	Change, middle-sized gap (points)	Change, large gap (points)
Pensioners, absolute	6.73	7.43	0.01	0.70	0.88	1.52
Pensioners, relative	10.62	11.37	0.03	0.75	0.66	0.93
Everyone, absolute	15.89	16.06	0.00	0.17	0.20	0.36
Everyone, relative	19.28	19.12	−0.01	−0.16	−0.03	−0.23

Note: Levels are on the OBR long-term path. Absolute: below 60% of the 2024-25 median, uprated by each path's CPI. Relative: below 60% of the median in the same run.

5.9 Example pensioners

Four hypothetical pensioners, run as households under each rule on the simulated path with a middle-sized gap, show how tax and benefits pass on a lower State Pension (Table 6, Figure 7); the OBR long-term run carries no example households. Each receives the full flat-rate State Pension and, unlike the survey runs, claims every benefit it is entitled to. A lower State Pension means less income tax for a pensioner above the personal allowance and more Pension Credit for one entitled to it; Pension Credit counts income after tax [State Pension Credit Regulations, 2002], and council tax reduction disregards the income of a pensioner receiving the guarantee credit [Council Tax Reduction Schemes (Prescribed Requirements) (England) Regulations, 2012]. The last row of Table 6 gives the share of the lower pension that reaches each pensioner's income in 2039-40.

Thresholds explain the differences, and two modelling conventions set the thresholds (Section 4). The Pension Credit guarantee rises with earnings under both rules, and the personal allowance stays at £12,570 to 2030-31 and then rises with CPI, so the full new State Pension is above the allowance and taxed in every year. With only the full new State Pension, a pensioner sits close to the guarantee after tax. Under the adjusted rule the pension (a) therefore receives more guarantee credit than under the triple lock in some years (2031-32, 2032-33, 2035-36 and 2039-40), and with it more council tax reduction. In those years more of the lower pension is made up, which is why the line in Figure 7 moves from year to year. The pensioner with a £15,000 private pension (b) is above every threshold and bears most of the loss, less income tax. The council tenant (c) is better off under the adjusted rule in 2039-40: guarantee credit brings maximum Housing Benefit, £1,334 a year more, and full council tax reduction, and together with less income tax these more than make up the lower pension. The older pensioner on the basic State Pension (d) is on the guarantee under both rules: extra guarantee credit makes up the lower pension in full, but they receive £272 less savings credit. Whether (a) and (c) reach the guarantee depends on both conventions: if the guarantee instead rose by the same percentage as the State Pension under each rule, or if the personal allowance were higher, they would be less likely to cross it.

Table 6: Example pensioners on the simulated path with a middle-sized gap in the final year: the change in each item of income under the adjusted Triple Lock, £ a year (a positive figure adds to income).

	(a)	(b)	(c)	(d)
State Pension	−£869	−£869	−£869	−£666
Income tax (less paid)	£174	£174	£174	£0
Pension Credit	£235	£0	£235	£394
Housing Benefit	£0	£0	£1,334	£0
Council tax reduction	£410	£0	£410	£0
Winter Fuel Payment	£0	£0	£0	£0
Income after tax and benefits	−£50	−£696	£1,284	−£272
Share of the State Pension fall that reaches income	6%	80%	−148%†	41%

Note: The last row is the change in income after tax and benefits as a share of the change in State Pension. † A negative share: income rises although the pension falls.
The examples: (a) single pensioner aged 70, State Pension only, owns their home; (b) single pensioner aged 70 with a £15,000 private pension, owns their home; (c) single pensioner aged 70, State Pension only, rents from the council at £120 a week; (d) single pensioner aged 90 on the old basic State Pension, rents from the council at £120 a week.

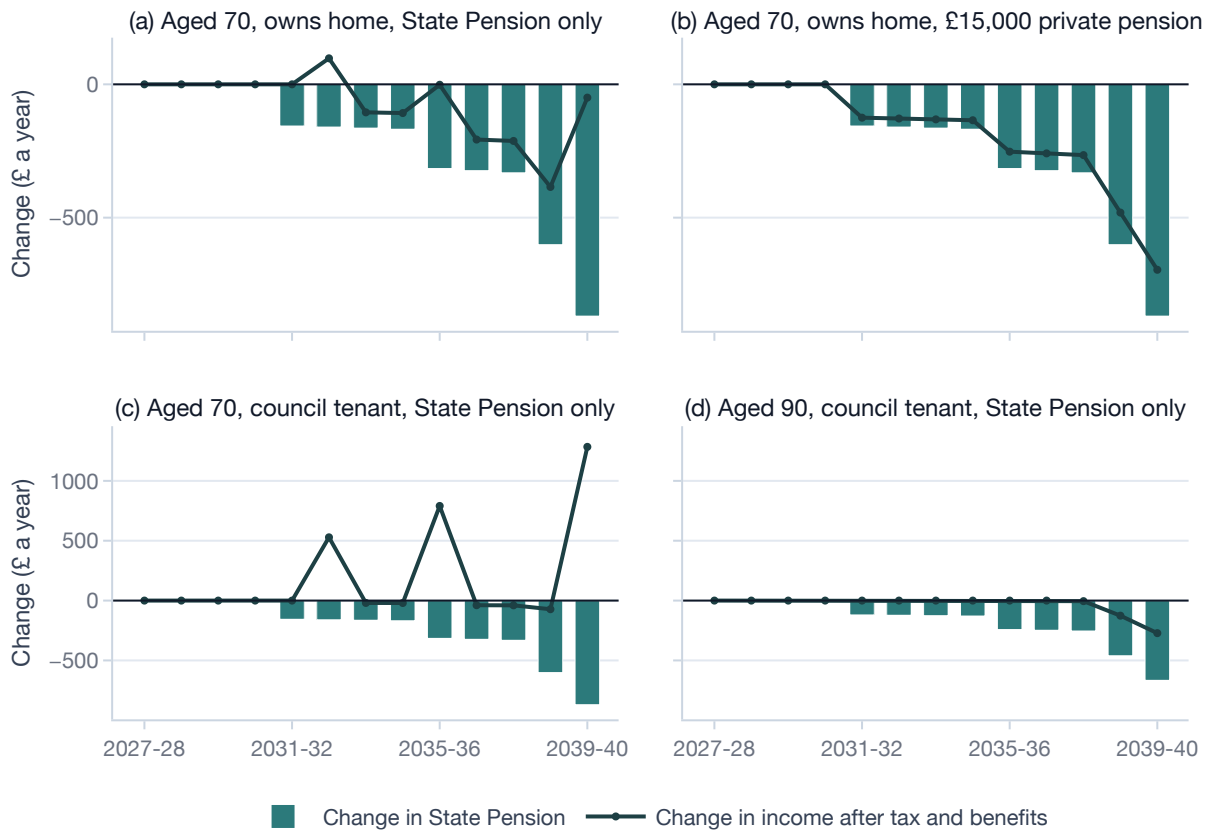


Figure 7: Benefits make up a varying share of the lower pension, depending on where each pensioner sits against the Pension Credit and other thresholds. The change in State Pension under the adjusted Triple Lock (bars) and in income after tax and benefits (line), £ a year, on the simulated path with a middle-sized gap.

6 Comparison with official costings

DWP costs the adjusted Triple Lock at £15bn in 2039-40 in cash terms [Department for Work and Pensions, 2026b], and the Budget repeats that figure (Section 1). It is direct spending for Great Britain, rounded to the nearest billion, from an uprating path that DWP does not publish, projected with its dynamic microsimulation model of the population and individual pension outcomes [see Li and O'Donoghue, 2013, on such models]; it leaves out tax and debt interest. Our nearest figures are the gross savings in Great Britain: £0.5bn on the OBR central forecast, £10.6bn on the OBR long-term path, and £11.1bn and £11.6bn on the two earnings scenarios. The comparison is not like for like. In answer to a parliamentary question, the government said that DWP's analysis did not use single-year earnings or prices growth assumptions [UK Parliament, 2026], so it does not correspond to any one of our paths. The population also differs: DWP projects deaths, migration and State Pension entitlements, while we reweight today's survey (Section 4.2), whose State Pension spending in Great Britain is 98% of DWP's forecast in 2026-27 (Table 8). DWP itself cautions against reading single-year point estimates from its figures, and its saving keeps growing, to £50bn in 2049-50; our horizon ends in 2039-40. DWP also gives £11bn in 2025-26 prices; our figures are in cash terms.

DWP also projects pensioner relative poverty after housing costs under the adjusted Triple Lock falling from around 14% to around 8% by 2049-50, against a poverty line projected forward with earnings. Our poverty rates (Section 5.6) are for 2039-40, use a line set each year from the simulated incomes, and compare the adjusted Triple Lock with the triple lock rather than with today, so they answer a different question.

Other published figures compare the triple lock with earnings uprating rather than with this rule, so they are context, not checks (Table 7). Our backward run, the adjusted Triple Lock from April 2012, is the nearest we have to those counterfactuals; it is a different rule from earnings uprating, since it keeps the price and floor protection.

Table 7: Other published figures beside our closest figure.

Source	Their figure	Our figure	What ours measures
Department for Work and Pensions [2026b]	£15bn in 2039-40; Great Britain, unpublished path	£0.5bn to £11.6bn	Gross saving, Great Britain, on the four scenarios
Office for Budget Responsibility [2026a]	2.5% under both rules in 2030-31 and 2031-32	2.5% in both years	The adjusted rule's rise in April 2030 and 2031, OBR central forecast
Office for Budget Responsibility [2026a]	Payments 1.4% lower on average than the triple lock, 1990–2025	5.1%	Level below the triple lock in 2026-27, rule from April 2017
Office for Budget Responsibility [2025]	Triple lock vs earnings link since 2012: £15.5bn by 2029-30	£7.8bn	Adjusted rule from April 2012, gross, 2026-27

Note: Our backward run (the adjusted rule from April 2012) keeps the protection against prices and the floor, so its saving is smaller than that from earnings uprating.

7 Discussion

Three points bear on how max-of-indices rules are costed. First, a central forecast understates the saving from removing a ratchet. On a smooth path on which earnings lead in almost every year the decomposition of Section 3.2 gives a gap near zero, while paths centred within 0.5 points of the same earnings growth, but varying from year to year, give a saving more than 23 times as large. Second, the OBR's long-term premium is a reduced-form allowance for the same effect. Applying it every year adds a lead for prices or the floor in every year, which no path of the inputs produces, so the OBR long-term path is a comparison with that costing convention, not a forecast. Third, the failed condition has a direction. The model expected earnings growth below the floor more often than it occurred, and such years are those in which $d_t > 0$; other things equal, that raises the number of changes of lead and so the simulated savings. A model that passed the test could give lower averages than the earnings scenarios here, and we cannot say by how much.

8 Conclusion

The adjusted Triple Lock keeps the triple lock's protection against prices and its 2.5% floor, but lets the pension fall back to its April 2029 value relative to earnings after years in which prices or the floor beat earnings. Its saving therefore depends on how often the lead passes between prices and earnings, not only on their expected growth. On the OBR's smooth central forecast it lowers State Pension spending by £0.5bn in 2039-40; on the OBR's own long-term assumption about the triple lock by £10.9bn; and averaged over simulated paths with earnings growth 0.5 points lower or higher by £11.4bn or £11.9bn, £7.8bn or £8.1bn after tax and benefits respond. These are scenarios, not a range with probabilities attached: our model of prices and earnings failed a registered test on past forecasts by overstating how often earnings growth falls below the floor, so the paper gives no expected value. DWP's £15bn for Great Britain rests on a path it does not publish and is not comparable like for like.

On every path about 30% of households, nearly all with a pensioner, have less income than under the triple lock, with Pension Credit and income tax making up part of the difference; on the OBR long-term path pensioner poverty after housing costs rises by 0.7 percentage points or less. The rule would also require a change to the statutory earnings link. The wider lesson applies to any uprating rule that takes the highest of several indices: its cost depends on how the indices vary from year to year, and costing it on a single smooth path understates the cost, as the OBR's long-term premium already recognises. Later releases will publish averages over the paths as labelled sensitivities, and a test on new outcomes can say whether they may be called estimates no earlier than October 2034.

The analysis has limits a reader should weigh. It gives no expected value, because the model that would weight the paths failed its registered test, and the simulated paths rest on one model of monthly prices and earnings with no periods of changing volatility. The population is reweighted to projected totals rather than aged: deaths, migration and contribution histories are not simulated, and the dataset's State Pension spending in Great Britain is 98% of DWP's forecast. Behaviour is held fixed, Housing Benefit and council tax reduction respond only for households already receiving them, and indirect taxes are left out, so the net figures count direct taxes and benefits only. After 2030 the central path follows the OBR's March 2026 long-term determinants, which the OBR describes as a projection, not a forecast, and the comparison with

DWP's figure is not like for like (Section 6).

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A Further results

This appendix gives the data against DWP's benefit expenditure and caseload forecasts [Department for Work and Pensions, 2026a] (Table 8) and the scores of each version of the model of prices and earnings in the test on past forecasts (Table 9).

Table 8: Our data against DWP's Spring Forecast 2026, Great Britain.

		Microcosm		Enhanced FRS 1.58.2	
2026-27	DWP	Value	% of DWP	Value	% of DWP
State Pension, excluding payments abroad (£bn)	148.3	144.8	98%	124.8	84%
Basic and new State Pension (a) (£bn)	131.0	123.1	94%	110.6	84%
State Pension recipients (m)	12.15	12.06	99%	11.07	91%
Pension Credit (£bn)	6.1	6.4	105%	7.0	116%
Pension Credit claims (m)	1.31	1.39	106%	1.46	112%
Housing Benefit, all ages (b) (£bn)	12.5	9.3	74%	8.2	66%
Housing Benefit, pension age (£bn)	7.3	7.2	99%	7.6	104%
Housing Benefit claims, pension age (m)	1.08	1.09	100%	1.11	103%

Note: Microcosm is the dataset every result uses; the Enhanced FRS is a check only. Claims are counted in benefit units. (a) DWP's figure includes payments abroad, which ours do not. (b) Excludes Universal Credit housing costs.

Table 9: The test on past forecasts, on the Budget's inputs, with April 2022 scored as in law.

	Seven forecasts		Thirteen forecasts				
Version	Yearly	Four-year	Yearly	Four-year	Floor bias	Score ratio	Verdict
Monthly (main version)	83%	5 of 7	90%	11 of 13	0.267	1.00	Fail
Monthly, two lags	67%	5 of 7	73%	11 of 13	0.243	1.07	Pass
Annual	79%	5 of 7	83%	11 of 13	0.271	1.15	Fail
Monthly, heavy tails	63%	5 of 7	71%	11 of 13	0.247	1.07	Pass
Monthly, normal shocks	83%	5 of 7	90%	11 of 13	0.275	1.03	Fail

Note: Seven forecasts: one a year, March 2016 to March 2022; thirteen forecasts: June 2010 to March 2022. Yearly: share of yearly gaps inside each version's central 80% band; four-year: four-year gaps inside it. Floor bias: average error in the share of years with earnings growth below the floor, seven-forecast test (limit in Table 1). Score ratio: the largest ratio of the version's accuracy scores to the main version's. The rule lets only the main version supply an average.